Los Angeles Indoor Air Quality Analysis

Executive Summary

This analysis examines the Indoor Air Quality Sensitivity Index across 2,495 census tracts in Los Angeles County to identify areas with higher sensitivity to indoor air quality conditions. The IAQ Sensitivity Index is measured by four different areas, Population Sensitivity, Outdoor Air Quality, Socioeconomic Status, and Housing Characteristics, with a maximum possible score of 100. The analysis focuses on the census tracts in the 90th percentile of the IAQ Sensitivity Score as they are considered extremely high. We examine the different components within the four areas that make up the IAQ Sensitivity Score to identify which contribute the most to these high scores.



Core Performance Synthesis
The data confirms a high IAQ Sensitivity Score for census tracts with high-levels of poverty, pollutants, renters, and health complications.

Critical Success Factors

- **Examining communities and cities with multiple elevated pollutant indicators.** Identify communities that have several pollutants at or above the 90th percentile cutoff to identify areas with multiple environmental exposure concerns.
- **Examining socioeconomic disadvantages.** Look into communities that have high IAQ Sensitivity Scores, poverty, educational disadvantage, and unemployment to identify overlapping socioeconomic vulnerabilities.
- **Analyze housing-related risks.** Examine the relationship between building age and children's lead risk along with renter percentage and unit size to understand housing-related vulnerability.
- **Consider health and demographic differences across communities.** Compare asthma, cardiovascular disease, and age-related indicators to understand how health vulnerabilities vary across communities.

Review Data & Findings: Insights and Implications

Pollutants and Outdoor Air Quality
Five pollutants make up the Outdoor Indoor Air Quality Score, ground-level ozone, PM2.5, toxic releases from facilities, diesel, and nitrogen dioxide. The score has a maximum of 20 points within the overall IAQ Sensitivity Index of 100.

The 90th-percentile cutoff was calculated for each pollutant's points and compared against the 246 communities in the highest 10% of IAQ Sensitivity Scores. Nitrogen dioxide was elevated the most for 99 communities, followed by PM2.5 in 78 communities, diesel in 42 communities, toxic releases from facilities in 31 communities and ground-level ozone in 1. In total, 166 of the 246 communities had at least one pollutant at or above its cutoff. Compton had four, making it a city with high exposure to pollutants.

Socioeconomic Disadvantage

Poverty, educational disadvantage and unemployment were examined to see whether socioeconomic disadvantage overlaps with high IAQ Sensitivity Scores.

Poverty contributes 7 points to the Socioeconomic Burden Score. The 90th-percentile cutoff was calculated for the entire dataset which was 64.59%. About 52.43% of the 246 high-sensitivity communities met the high-poverty classification. Los Angeles had the most prominent representation, with Vermont Vista considered a high-poverty community. Along with this it was found that from the cities/communities that have at least 1 pollutant at a high level (166 out of 246), 89 were also in high poverty.

In Los Angeles County, Indoor air quality sensitivity (IAQ) and socioeconomic disadvantages such as lower educational attainment and unemployment overlap across communities. After cleaning up the dataset, Excel pivot tables and visualizations(charts) compared the three factors: educational disadvantage, unemployment, and IAQ sensitivity, through percentile rankings and underlying rates. Educational disadvantage is defined as any adult over the age of 25 without a high school diploma. The data show that communities with higher educational disadvantage or unemployment often have high IAQ sensitivity rankings, yet the relationship is not consistent across communities. From the data, three communities showed the highest ranking across the measures: Vermont Vista, Willowbrook, and Green Meadows. In this specific set, Lancaster demonstrated a mixed pattern. Lancaster displayed a high IAQ sensitivity and unemployment rankings but ranked lower in the educational disadvantage section. The unique output of results illustrates why having multiple indicators is valuable and should always be considered when examining community vulnerability.

The analysis concludes by suggesting that socioeconomic disadvantages and IAQ sensitivity can overlap, but it doesn't mean that correlation equals causation. There is some vulnerability seen from the IAQ Sensitivity Index, rather than directly measuring indoor pollutant concentrations. Overall, the findings and analysis showcase that further recommendations are needed to improve conditions such as community planning, equity-informed resource allocation, indoor air quality assessments, and the continued monitoring of environmental and socioeconomic conditions. For the sake of better understanding local conditions, future research should utilize historical trends. More structured housing data, geographical mapping, and community forum input will be required to help public and community organizations better find areas that may need investigation.

Housing Characteristics

Housing Characteristics were split into 4 categories, Children's Lead Risk, Renter %, Unit Size, and Building Age.

When analyzing the data between **Children's Lead Risk** and **Building Age**, the data shows a strong positive relationship across LA County Communities. Older housing stock, specifically homes built on or before 1979, is associated with elevated lead exposure risk for children due to the potential presence of lead-based paint. SPA 6 (South) and SPA 7 (East) have the highest reported children's lead risk scores, at 7.02 and 7.00 respectively. This pattern suggests the age of residential buildings is an important indicator of potential lead exposure, though the relationship does not establish that building age alone causes higher risk.

Taking a look at **Renter %**, we utilized the comparison of the overall housing score and indoor air quality sensitivity to find a general pattern. Communities with higher renter percentages tend to have higher housing characteristics scores. This suggests that renter prevalence may be associated with housing conditions that contribute to indoor air quality sensitivity. However, the data shows considerable variation, indicating that renter percentage alone does not fully explain differences in indoor air quality sensitivity across communities.

Finally, analyzing **Average Unit Size** reveals a negative relationship between average unit size and indoor air quality sensitivity scores. Communities with smaller average units generally tend to have higher sensitivity scores, while those with larger units tend to have lower scores. This pattern may reflect differences in housing density, ventilation, or other building characteristics.

These findings suggest that IAQ Sensitivity across LA County can be shaped by interconnected housing characteristics. Building age is closely associated with children's lead risk, renter percentage shows a general relationship with housing characteristics scores, and smaller average unit sizes are associated with higher IAQ Sensitivity. Together, these patterns highlight the importance of considering both physical characteristics of housing and populations living in it when assessing environmental health risks.

Population Health Sensitivity

Asthma-Related Health Vulnerability

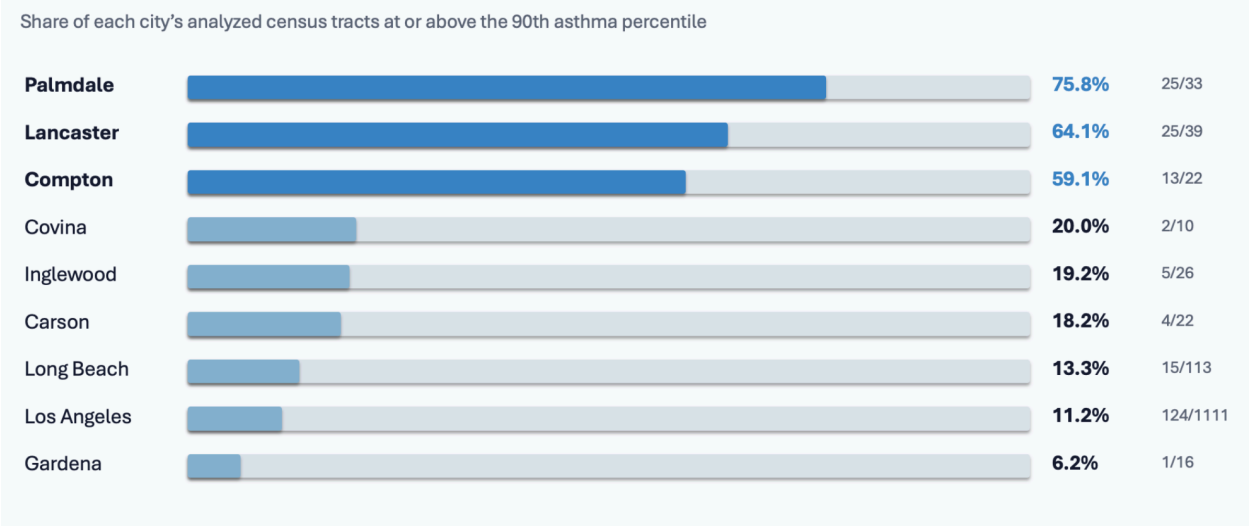
Asthma-related health vulnerability was examined to understand how respiratory health indicators differ across Los Angeles County communities and how these indicators overlap with potentially vulnerable age groups. The analysis focused on three questions:

1. Which cities have the highest concentration of high-asthma census tracts?
2. Where high asthma rankings overlap with populations of young children and older adults?
3. How do asthma rankings differ among census tracts within the same city?

The analysis used the Asthma Percentile indicator from the Los Angeles County Indoor Air Quality Sensitivity Index, Version 1.0. Census tracts were classified as having high asthma-related rankings when their asthma percentile was at or above 90. All areas labeled unincorporated were excluded.

Unlike the overall IAQ Sensitivity Score, which combines multiple environmental, demographic, socioeconomic, and housing indicators, the Asthma Percentile specifically measures a census tract's relative ranking on the dataset's asthma-related health indicator. These rankings should not be interpreted as the percentage of residents diagnosed with asthma.

Geographic Concentration of High-Asthma Census Tracts
The first analysis examined the proportion of census tracts within each incorporated city that ranked at or above the 90th percentile for asthma. Comparing proportions rather than total counts allowed cities with different numbers of census tracts to be evaluated more consistently. Three cities demonstrated the highest concentrations:



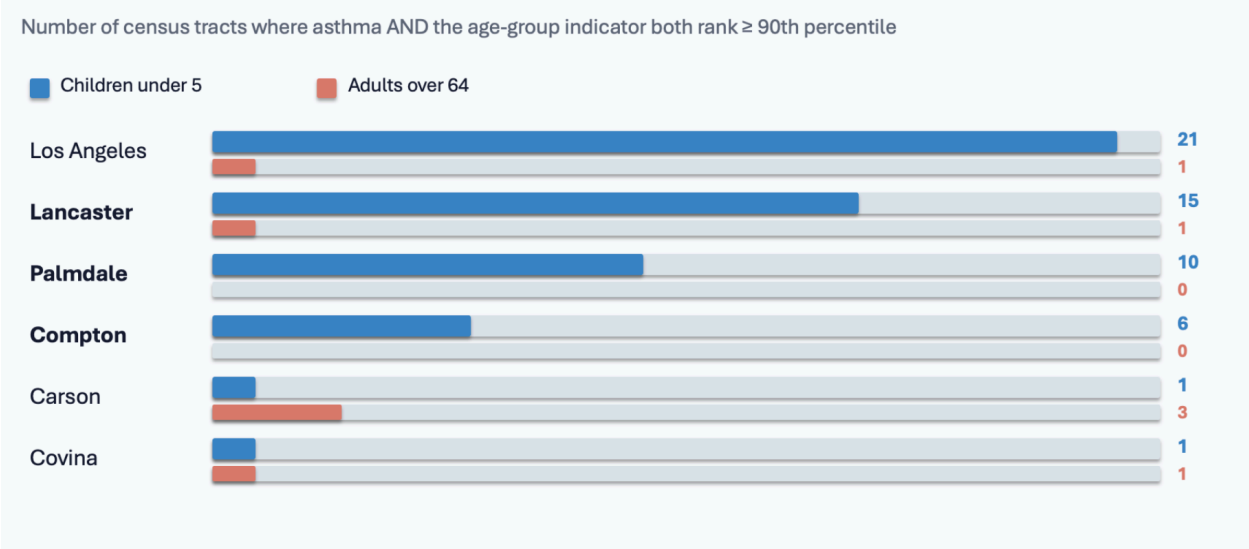
Palmdale had the highest proportion, with approximately three out of every four analyzed census tracts meeting the high-asthma threshold. Lancaster followed at 64.1%, while Compton had 59.1%. Other cities demonstrated substantially lower proportions. Covina had 20.0%, Inglewood had 19.2%, Carson had 18.2%, Long Beach had 13.3%, Los Angeles had 11.2%, and Gardena had 6.2%.

An important finding was the difference between absolute counts and proportions. Although Los Angeles contained more qualifying high-asthma census tracts overall, its much larger number of census tracts resulted in a lower percentage than Palmdale, Lancaster, or Compton.
This demonstrates why a city with the largest number of high-ranking areas is not necessarily the city with the greatest geographic concentration of those areas.

Key finding: Palmdale, Lancaster, and Compton had the highest proportions of analyzed census tracts with asthma-related rankings at or above the 90th percentile. This indicates that high asthma rankings are more geographically widespread across these cities than in the other incorporated cities included in the comparison.

Geographic Overlap With Vulnerable Age Groups
The second analysis examined whether high asthma rankings occurred in the same census tracts as high population rankings for children under age 5 and adults over age 64.
Both age groups are relevant to respiratory health vulnerability because they may be more susceptible to certain effects of air pollution. However, the dataset does not provide age-specific asthma prevalence rates.

To identify geographic overlap, census tracts were selected when both the Asthma Percentile and the corresponding age-group percentile reached at least 90.
The final visualization highlighted Los Angeles, Lancaster, Carson, and Covina.



The results showed that Los Angeles had 21 census tracts where high asthma rankings overlapped with high rankings for children under 5. Lancaster followed with 15.
This was particularly noteworthy because Lancaster had already appeared among the cities with the highest proportions of high-asthma census tracts in the first analysis. Its appearance in both analyses suggests that some of its high-ranking asthma areas also contain relatively high rankings for young-child populations.

Palmdale and Compton, the other two leading cities from the first analysis, also demonstrated this geographic overlap. Palmdale had 10 census tracts where both indicators reached the 90th percentile, while Compton had six.

Carson showed a different demographic pattern. Three of its high-asthma census tracts also ranked highly for adults over age 64, compared with one tract for children under 5. Covina had one qualifying census tract in each age-group category.
These differences indicate that high-asthma communities do not necessarily share the same age-related population characteristics.
The analysis also demonstrated why the distinction between geographic overlap and individual health outcomes is important. A census tract with high asthma and high young-child population

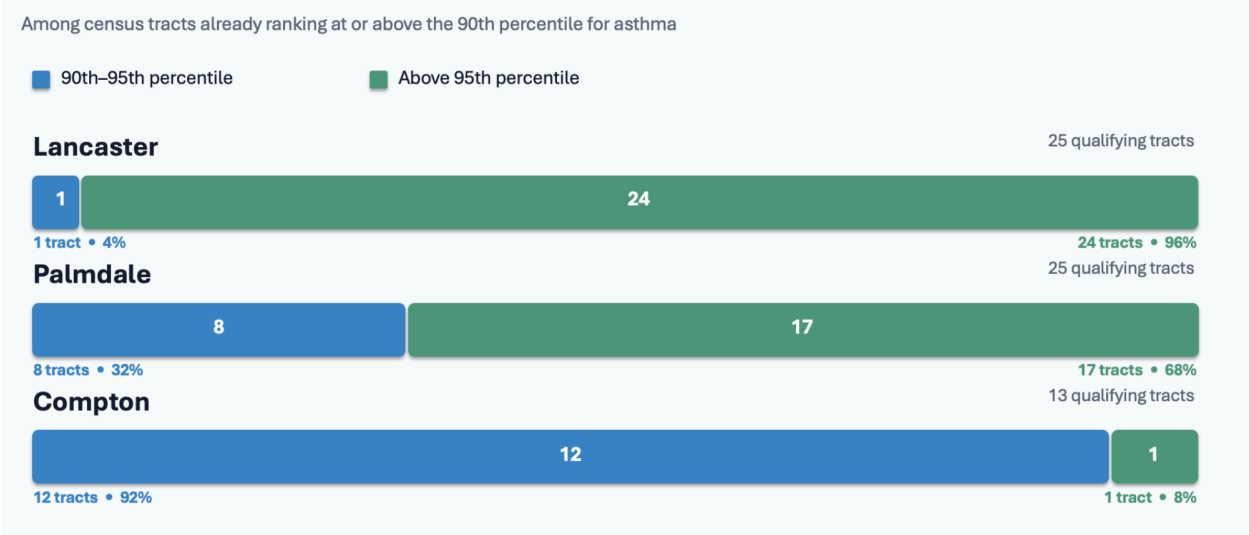
rankings does not establish that the children living in that tract have asthma. Rather, it identifies an area where two potentially relevant vulnerability indicators occur together.

Key finding: High asthma rankings overlap with high young-child population rankings in several communities, particularly Los Angeles and Lancaster. Carson demonstrates a different pattern, with greater overlap involving older adults. These findings highlight the importance of considering demographic differences when examining community health vulnerability.

Differences in Asthma Rankings Within High-Asthma Cities

The third analysis investigated whether census tracts already meeting the 90th-percentile asthma threshold had similar rankings or whether some cities contained a greater concentration of even higher-ranking census tracts. This comparison focused on Palmdale, Lancaster, and Compton, the three cities identified in the first analysis.

Qualifying census tracts were divided into two categories: those ranking from the 90th through the 95th percentile and those ranking above the 95th percentile.



Lancaster demonstrated the greatest concentration of rankings above the 95th percentile. Of its 25 qualifying high-asthma census tracts, 24 ranked above the 95th percentile, representing 96%.

Palmdale also had a substantial concentration of very high rankings, with 17 of its 25 qualifying census tracts, or 68%, above the 95th percentile.

Compton displayed a different pattern. Although 13 census tracts met the original 90th-percentile threshold, 12 were ranked between the 90th and 95th percentiles, while only one ranked above the 95th percentile.

These findings reveal an important distinction between the overall concentration of high-asthma census tracts and the distribution of rankings within that group.

For example, Palmdale had the highest proportion of all analyzed census tracts meeting the 90th-percentile threshold, but Lancaster had the highest proportion of its qualifying census tracts ranking above the 95th percentile.

This indicates that the geographic concentration of high asthma rankings and the relative ranking levels within those areas are two different characteristics of community vulnerability. It is important to note that a census tract at the 99th percentile does not necessarily have a proportionally greater number of asthma cases than a tract at the 91st percentile. Percentiles indicate relative ranking rather than the magnitude of the difference in asthma prevalence.

Key finding: Lancaster had the highest proportion of its qualifying high-asthma census tracts above the 95th percentile, followed by Palmdale and Compton. This demonstrates that even communities with high concentrations of asthma-related indicators can differ substantially in how their census tracts rank.

Overall Insights and Implications

Together, the three analyses reveal that asthma-related health vulnerability is not geographically uniform across Los Angeles County.

First, Palmdale, Lancaster, and Compton stand out for having the highest proportions of analyzed census tracts meeting the 90th-percentile asthma threshold.

Second, high asthma rankings overlap with different age-related population characteristics across communities. Los Angeles and Lancaster show notable overlap with high young-child population rankings, while Carson demonstrates a comparatively greater overlap with the older-adult indicator.

Third, even among census tracts already identified as high-ranking, the distribution of asthma percentiles differs between cities. Lancaster's qualifying census tracts are particularly concentrated above the 95th percentile, while Compton's qualifying tracts are primarily concentrated between the 90th and 95th percentiles.

These findings illustrate the value of census tract-level analysis. Citywide totals alone can obscure differences in geographic concentration, demographic overlap, and the relative ranking of individual census tracts. The results also complement the broader Indoor Air Quality Sensitivity Index analysis by adding a health-related perspective to the environmental, socioeconomic, and housing patterns examined in other sections of this report.

However, the analysis is descriptive and does not establish that indoor air quality conditions caused the observed asthma-related rankings. The dataset's asthma indicator is a measure of relative health-related ranking, not a direct measurement of indoor pollutant exposure or age-specific asthma prevalence.

Overall, the findings suggest that understanding asthma-related vulnerability requires examining not only which cities have high-ranking census tracts, but also how those rankings are distributed and which population characteristics overlap geographically with them. This information provides a foundation for more targeted investigation of respiratory health concerns and indoor air quality conditions across Los Angeles County.

Strategic Recommendations: Ideas and Potential Impact

1.) Reduce Pollution Exposure

Prioritize communities with multiple elevated pollutant indicators. Increase air-quality monitoring, investigate pollution sources, and improve home air filtration. Organize community-led air quality forums with residents, neighborhood councils, and nonprofits to gather concerns about indoor air quality and nearby pollution sources.

2.) Improve Housing Conditions

Develop a program to handle housing hazards in communities with older housing and elevated children's lead risk. Inspect for housing hazards such as lead, mold, and improve ventilation to control moisture.

Next Steps: Inspirations and Inquiries

To move from describing patterns in this analysis to guiding decisions we must investigate specific data anomalies.

Strategic Inquiries

- **What explains the multi-pollutant areas?** Determine why 63 communities have more than 1 high pollutant and why the city of Compton has 4 high pollutants.
- **What else drives health issues?** Test other factors that may lead to high levels of asthma or cardiovascular disease other than age and pollutants.

Operational Inspirations

- **Linguistic isolation and healthcare access:** which communities with linguistic isolation are more susceptible to being uninsured.
- **Overlapping disadvantage:** communities that rank high for pollutants, poverty, education, and unemployment together.
- **Housing and pollutants:** the correlation between unit size, renter percentage, and pollutant levels.

Data Quality & Tracking Needs

- **Complete Component Coverage:** We must complete the analysis on linguistic isolation and show how it can contribute to socioeconomic score.

Final Summary

The primary objective in assessing the Indoor Air Quality Sensitivity Score across Los Angeles County is to focus on communities where pollution, poverty, educational disadvantage and healthcare access overlap.